

Chapter 2

One-dimensional issues

This chapter is devoted to the special features of the 1D case, and to a less known transport map in the higher-dimensional case, called the *Knothe map*, built from 1D constructions and related to some degenerate transport cost. The discussion section deals with applications of 1D transport maps to image processing and geometric inequalities.

2.1 Monotone transport maps and plans in 1D

We already discussed, in Section 1.3, the quadratic cost in 1D: as soon as the source measure μ has no atoms, there exists an optimal map, which is monotone non-decreasing.

In this section we want to discuss monotone (in the following “monotone” means “monotone non-decreasing”) transport maps between two given measures in terms of their cumulative distribution functions, and generalize to the case where μ may have atoms. This discussion is independent of optimal transport considerations.

Box 2.1. – Memo – Cumulative distribution function

Definition - Given a probability measure $\mu \in \mathcal{P}(\mathbb{R})$ we define its cumulative distribution function F_μ through

$$F_\mu(x) = \mu((-\infty, x]).$$

The cumulative distribution function F_μ is easily seen to be non-decreasing and right-continuous since if $x_n \rightarrow x$ with $x_n > x$, then $(-\infty, x] = \cap_n (-\infty, x_n]$ and hence $\mu((-\infty, x]) = \lim_n \mu((-\infty, x_n]) = \inf_n \mu((-\infty, x_n])$. It is continuous at any point where μ has no atom since if $x_n \rightarrow x$ with $x_n < x$, then $\lim_n \mu((-\infty, x_n]) = \mu((-\infty, x])$.

The CDF F_μ is important because it characterizes the measure μ . Indeed, the sets of the form $(-\infty, x]$ are enough to generate all open sets: we have $\mu(]a, b]) = F_\mu(b) - F_\mu(a)$, $\mu(]a, b]) = \sup\{F_\mu(t) - F_\mu(a) : t < b\}$ and the measure of any open set is obtained via countable disjoint unions of open intervals.

In $\mathbb{R}^d$ the situation is trickier, but the knowledge of $\mu(]-\infty, x_1] \times \cdots \times]-\infty, x_d])$ for every $x = (x_1, \dots, x_d)$ is enough, as in 1D, to characterize μ (by finite differences one could

get all the semi-open rectangles $]x_1^-, x_1^+] \times \cdots \times]x_d^-, x_d^+]$, and by countable disjoint unions all the open sets).

Unfortunately, the CDF above cannot always be inverted, as it is not always strictly increasing, but we can define a pseudo-inverse:

Definition 2.1. Given a non-decreasing and right-continuous function $F : \mathbb{R} \rightarrow [0, 1]$, its pseudo-inverse is the function $F^{[-1]} : [0, 1] \rightarrow \mathbb{R}$ given by

$$F^{[-1]}(x) := \inf\{t \in \mathbb{R} : F(t) \geq x\},$$

where the infimum is a minimum as soon as the set is non-empty (otherwise it is $+\infty$) and bounded from below (otherwise it is $-\infty$), thanks to right-continuity of F .

Note, as a simple consequence of the definition of pseudo-inverse, that we have

$$F^{[-1]}(x) \leq a \Leftrightarrow F(a) \geq x; \quad F^{[-1]}(x) > a \Leftrightarrow F(a) < x. \quad (2.1)$$

We look now at some properties of the pseudo-inverse of cumulative distribution functions.

Proposition 2.2. If $\mu \in \mathcal{P}(\mathbb{R})$ and $F_\mu^{[-1]}$ is the pseudo-inverse of its cumulative distribution function F_μ , then $(F_\mu^{[-1]})_\#(\mathcal{L}^1 \llcorner [0, 1]) = \mu$.

Moreover, given $\mu, \nu \in \mathcal{P}(\mathbb{R})$, if we set $\eta := (F_\mu^{[-1]}, F_\nu^{[-1]})_\#(\mathcal{L}^1 \llcorner [0, 1])$, then $\eta \in \Pi(\mu, \nu)$ and $\eta((-\infty, a] \times (-\infty, b]) = F_\mu(a) \wedge F_\nu(b)$.

Proof. For the first part of the statement, using (2.1), we write

$$|\{x \in [0, 1] : F_\mu^{[-1]}(x) \leq a\}| = |\{x \in [0, 1] : F_\mu(a) \geq x\}| = F_\mu(a)$$

and this proves that the image measure is μ , using the characterization of a measure through its CFD (see the Memo box 2.1 above).

For the second part of the statement, $\eta \in \Pi(\mu, \nu)$ is just a consequence of the first. Then, let us compute

$$\begin{aligned} \eta((-\infty, a] \times (-\infty, b]) &= |\{x \in [0, 1] : F_\mu^{[-1]}(x) \leq a, F_\nu^{[-1]}(x) \leq b\}| \\ &= |\{x \in [0, 1] : F_\mu(a) \geq x, F_\nu(b) \geq x\}| = F_\mu(a) \wedge F_\nu(b), \end{aligned}$$

which is the desired equality. □ □

Definition 2.3. We will call the transport plan $\eta := (F_\mu^{[-1]}, F_\nu^{[-1]})_\#(\mathcal{L}^1 \llcorner [0, 1])$ the co-monotone transport plan between μ and ν , and denote it by γ_{mon} .

Now, consider two measures $\mu, \nu \in \mathcal{P}(\mathbb{R})$: we want to build a monotone transport map (and not only a plan) sending μ onto ν , provided μ is atomless.

We first need a simple lemma on the CFD of atomless measures.

Lemma 2.4. *If $\mu \in \mathcal{P}(\mathbb{R})$ is atomless, then $(F_\mu)_\# \mu = \mathcal{L}^1 \llcorner [0, 1]$. As a consequence, for every $\ell \in [0, 1]$ the set $\{x : F_\mu(x) = \ell\}$ is μ -negligible.*

Proof. First note that F_μ is continuous because μ is atomless. Hence, for $a \in]0, 1[$, the set $\{x : F_\mu(x) \leq a\}$ is a closed interval of the form $]-\infty, x_a]$, with $F_\mu(x_a) = a$. Hence, $\mu(\{x : F_\mu(x) \leq a\}) = F_\mu(x_a) = a$, which proves $(F_\mu)_\# \mu = \mathcal{L}^1 \llcorner [0, 1]$.

As a consequence, for $\ell \in [0, 1]$, the sets $\{x : F_\mu(x) = \ell\}$ are μ -negligible, since otherwise the image measure $(F_\mu)_\# \mu$ would have an atom at ℓ , which contradicts the first part of the statement. $\square$

Theorem 2.5. *Given $\mu, \nu \in \mathcal{P}(\mathbb{R})$, suppose that μ is atomless. Then, there exists unique a non-decreasing map $T_{\text{mon}} : \mathbb{R} \rightarrow \mathbb{R}$ such that $(T_{\text{mon}})_\# \mu = \nu$.*

Proof. First, let us build one such a map. Let us consider the cumulative distribution functions F_μ and F_ν , and define the map T_{mon} through

$$T_{\text{mon}}(x) := F_\nu^{[-1]}(F_\mu(x)).$$

This quantity is well-defined and belongs to $\mathbb{R}$ provided $F_\mu(x) \in]0, 1[$ (so that the set on which we take the infimum in Definition 2.1 is neither empty nor unbounded from below). The sets $\{x : F_\mu(x) = 0\}$ and $\{x : F_\mu(x) = 1\}$ are μ -negligible thanks to Lemma 2.4. Hence, T_{mon} is well-defined μ -a.e.

The fact that T_{mon} is monotone non-decreasing is obvious; we just have to prove $(T_{\text{mon}})_\# \mu = \nu$. Since we already know $(F_\nu^{[-1]})_\#(\mathcal{L}^1 \llcorner [0, 1]) = \nu$, by composition we just need to use Lemma 2.4, which proves $(F_\mu)_\# \mu = \mathcal{L}^1 \llcorner [0, 1]$.

We now pass to the proof of uniqueness. Consider any non-decreasing function T such that $T_\# \mu = \nu$.

From monotonicity we have $T^{-1}(]-\infty, T(x)]) \supset]-\infty, x]$. We deduce

$$F_\mu(x) = \mu(]-\infty, x]) \leq \mu(T^{-1}(]-\infty, T(x)])) = \nu(]-\infty, T(x)]) = F_\nu(T(x)),$$

which means $T(x) \geq F_\nu^{[-1]}(F_\mu(x))$. Suppose that the inequality is strict. This means that there exists $\varepsilon_0 > 0$ such that $F_\nu(T(x) - \varepsilon) \geq F_\mu(x)$ for every $\varepsilon \in]0, \varepsilon_0[$. On the other hand, from $T^{-1}(]-\infty, T(x) - \varepsilon]) \subset]-\infty, x[$, we get $F_\nu(T(x) - \varepsilon) \leq F_\mu(x)$.

Hence, we obtain $F_\nu(T(x) - \varepsilon) = F_\mu(x)$ for every $\varepsilon \in]0, \varepsilon_0[$. Thus, $F_\nu(T(x) - \varepsilon)$ is the value that F_ν takes on an interval where it is constant. These intervals are a countable quantity, hence the values of F_ν on those intervals are also countable. We call ℓ_i these values. As a consequence, the points x where $T(x) > F_\nu^{[-1]}(F_\mu(x))$ are contained in $\bigcup_i \{x : F_\mu(x) = \ell_i\}$. Lemma 2.4 proves that this set is negligible. This allows to conclude $T(x) = F_\nu^{[-1]}(F_\mu(x))$ μ -a.e. $\square$

Remark 2.6. Note that the previous proof was complicated by the possibility that the cumulative distribution functions could be either discontinuous or not strictly increasing. Should F_ν be continuous and strictly monotone (which means that ν is atomless and supported on the whole $\mathbb{R}$), then one would simply have

$$T_{\text{mon}} = (F_\nu)^{-1} \circ F_\mu.$$

Remark 2.7. As a consequence of the explicit formula of the previous remark, one can also study the regularity of this map T_{mon} depending on the regularity of μ and ν . Indeed, as soon as these measures are fully supported and have no atoms, the two functions F_μ and F_ν are homeomorphisms, and so is T_{mon} . Moreover, if μ and ν are absolutely continuous, with continuous densities which do not vanish, then they are also diffeomorphisms, and so is T_{mon} . In general, the regularity of T_{mon} is one degree higher than that of the two measures, as it is the case for F_μ, F_ν, F_μ^{-1} and F_ν^{-1} .

In the next section we will see that the map T_{mon} that we have just built optimizes a whole class of transport costs. To prove it, we will need the following characterization of γ_{mon} and T_{mon} .

Lemma 2.8. *Let $\gamma \in \Pi(\mu, \nu)$ be a transport plan between two measures $\mu, \nu \in \mathcal{P}(\mathbb{R})$. Suppose that it satisfies the property*

$$(x, y), (x', y') \in \text{spt}(\gamma), x < x' \Rightarrow y \leq y'. \quad (2.2)$$

Then, we have $\gamma = \gamma_{\text{mon}}$. In particular, there is a unique γ satisfying (2.2).

Moreover, if μ is atomless, then $\gamma = \gamma_{\Gamma_{\text{mon}}}$.

Proof. For the first part of the statement we just need to prove

$$\gamma((-\infty, a] \times (-\infty, b]) = F_\mu(a) \wedge F_\nu(b).$$

Indeed, from Proposition 2.2, this condition is satisfied by γ_{mon} , and we saw that this is enough to characterize a measure on $\mathbb{R}^2$.

Consider the two sets $A =]-\infty, a] \times]b, +\infty[$ and $B =]a, +\infty[\times]-\infty, b]$. From the assumption on γ , it is not possible to have both $\gamma(A) > 0$ and $\gamma(B) > 0$ (otherwise we would have points in $\text{spt}(\gamma)$ violating the condition (2.2)). Hence, we can write

$$\gamma((-\infty, a] \times (-\infty, b]) = \gamma(((-\infty, a] \times (-\infty, b]) \cup A) \wedge \gamma(((-\infty, a] \times (-\infty, b]) \cup B).$$

Yet, we have

$$\gamma(((-\infty, a] \times (-\infty, b]) \cup A) = \gamma((-\infty, a] \times \mathbb{R}) = F_\mu(a)$$

and

$$\gamma(((-\infty, a] \times (-\infty, b]) \cup B) = \gamma(\mathbb{R} \times (-\infty, b]) = F_\nu(b).$$

This proves the first part of the claim. For the second part, we suppose μ to be atomless. For any point $x \in \mathbb{R}$ one can define the interval I_x as the minimal interval I such that $\text{spt}(\gamma) \cap (\{x\} \times \mathbb{R}) \subset \{x\} \times I$. This interval can obviously be reduced to a singleton. The assumption on γ implies that the interior of all these intervals are disjoint (and ordered). In particular, there can be at most a countable quantity of points such that I_x is not a singleton. Since these points are μ -negligible (as a consequence of μ being atomless), we can define μ -a.e. a map T such that γ is concentrated on the graph of T . This map will be monotone non-decreasing because of (2.2), and this gives $T = T_{\text{mon}}$ since we already know the uniqueness of a non-decreasing map with fixed marginals (Theorem 2.5). $\square$ $\square$

2.2 The optimality of the monotone map

Now that we know quite well the properties, definitions and characterizations of the map T_{mon} and of the plan γ_{mon} , we can see that they are, in the 1D case, optimal for several different costs, and not only for the quadratic one. This is really specific to the one-dimensional case, it will not be true in higher dimension. The cost that we will consider will be convex functions of the difference $x - y$ and, to stress the possible asymmetric behavior of these costs, we prefer to write $h(y - x)$ instead of $h(x - y)$.

Theorem 2.9. *Let $h : \mathbb{R} \rightarrow \mathbb{R}_+$ be a strictly convex function, and $\mu, \nu \in \mathcal{P}(\mathbb{R})$ be probability measures. Consider the cost $c(x, y) = h(y - x)$ and suppose that (KP) has a finite value. Then, (KP) has a unique solution, which is given by γ_{mon} . In the case where μ is atomless, this optimal plan is induced by the map T_{mon} .*

Moreover, if the strict convexity assumption is withdrawn and h is only convex, then the same γ_{mon} is actually an optimal transport plan, but no uniqueness is guaranteed anymore.

Proof. We will use the fact that the support of any optimal γ is a c -CM set Γ . This means in particular that for $(x, y), (x', y') \in \Gamma$ we have

$$h(y - x) + h(y' - x') \leq h(y' - x) + h(y - x'). \quad (2.3)$$

We only need to show that this implies (in the strictly convex case) a monotone behavior : we will actually deduce from (2.3) that $x < x'$ implies $y \leq y'$, and this will allow to conclude from Lemma 2.8.

To prove $y \leq y'$ suppose by contradiction $y > y'$ and denote $a = y - x$, $b = y' - x'$ and $\delta = x' - x > 0$. Condition (2.3) reads $h(a) + h(b) \leq h(b + \delta) + h(a - \delta)$. Moreover, the assumption $y' < y$ implies $b + \delta < a$. Hence, $b + \delta$ and $a - \delta$ are located in the segment between b and a (and $b < a$). More precisely, we have

$$b + \delta = (1 - t)b + ta, \quad a - \delta = tb + (1 - t)a, \quad \text{for } t = \frac{\delta}{a - b} \in]0, 1[.$$

Thus, strict convexity yields

$$\begin{aligned} h(a) + h(b) &\leq h(b + \delta) + h(a - \delta) \\ &< (1 - t)h(b) + th(a) + th(b) + (1 - t)h(a) = h(a) + h(b). \end{aligned}$$

This gives a contradiction and proves the statement in the strictly convex case.

The statement when h is only convex is trivial if h is constant (since every γ is optimal) and, if not, it is obtained by approximation. Lemma 2.10 below proves that there exist a sequence of strictly convex functions h_ε such that $h \leq h_\varepsilon \leq (1 + \varepsilon)h + \varepsilon$. Let us take the transport cost $c_\varepsilon(x, y) := h_\varepsilon(y - x)$. In this case we know that γ_{mon} optimizes the cost $\int c_\varepsilon d\gamma$ and hence

$$\int h(y - x) d\gamma_{\text{mon}} \leq \int h_\varepsilon(y - x) d\gamma_{\text{mon}} \leq \int h_\varepsilon(y - x) d\gamma \leq \varepsilon + (1 + \varepsilon) \int h(y - x) d\gamma$$

for all $\gamma \in \Pi(\mu, \nu)$. Passing to the limit as $\varepsilon \rightarrow 0$, we get that γ_{mon} also optimizes the cost c . $\square$ $\square$